📑 Product Requirements Document: MTU Centralized Lost and Found Portal

1. 🎯 Product Overview

Product VisionTo transform campus property recovery from a manual, fragmented process into a high-trust, automated digital ecosystem. The vision is to replace outdated physical notice boards with a 24/7 accessible digital hub that minimizes the psychological stress of property loss.

Problem Statement

Fragmented Information: Current reliance on word-of-mouth or static notice boards is unreliable and easily overlooked.

Manual Delays: Human administrative processes are time-consuming and prone to inconsistencies during peak periods.

Low Recovery Rates: Without a centralized database, items remain unclaimed in security offices, leading to permanent loss.

Inefficient Tradition: Manual desks fail to meet the modern expectations of a digital university.

The Solution

A centralized, web-based portal designed specifically for Mountain Top University to monitor, report, and track lost and found items through automated matching and secure owner verification.

2. 🚀 Goals & Success Metrics

Strategic Goals

Analyze & Automate: Move away from manual recovery processes through digital reporting.

Seamless Tracking: Develop a user-friendly system for reporting and tracking items in real-time.

Secure Verification: Integrate automated matching and secure owner verification to prevent theft.

Operational Efficiency: Minimize time spent by security staff on routine inquiries.

Key Performance Indicators (KPIs)

Recovery Speed: Reduce item recovery time from days to hours via instant notifications.

Claim Accuracy: Achieve "Zero Wrongful Claims" by eliminating theft through multi-step verification.

Success Rate: Increase the percentage of returned high-value items like Laptops and iPads.

Community Anxiety: Measurably reduce student anxiety by providing a reliable recovery system.

3. 👥 User Personas

PersonaNeeds

Student / StaffNeeds to report lost items quickly and browse found items without exposing sensitive details.

Finder (Guest/Student)Needs a simple interface to report found property without administrative friction.

MTU Security (Admin)Needs to verify ownership proof, manage physical inventory, and oversee the hand-over process.

4. 🛠️ Core Features (The "Build" Phase)

A. Reporting System

Lost Item Form: User-friendly wizard to log category, last seen location, and description.

Found Item Form: Simple interface for finders to upload item info to the database.

Image Upload: Basic capability to attach one photo per report to provide visual context.

B. The Centralized Database

Search & Filter: Users can browse a public ledger of found items (redacted for security).

Manual Matching: A "This looks like my item" button to trigger a claim request.

Smart Match Engine: Automated service flagging similarities between "Lost" and "Found"

reports.

C. Verification Gate

Claim Management: Manual multi-step verification where owners provide specific "hidden" details to prove ownership.

Secure Verification: Integration of automated matching and secure owner verification.

5. 🔄 Navigation & Screen Flow

User Flow (Lost/Found)

Auth: Login via MTU Institutional Email.

Dashboard: View active reports and potential matches.

Reporting Wizard: Select "Lost" or "Found" > Category > Details > Location > Image.

Public Ledger: Browse redacted "Found" items with search/filter.

Claim Portal: Submit "Hidden Details" and Proof of Ownership (Receipts/Photos).

Admin Flow (Security Staff)

Operational Dashboard: Overview of total items reported vs. recovered.

Claim Review: Compare user-submitted proof against the physical item in custody.

Inventory Management: Update item status (Active > Claimed > Archived).

6. 🏗️ Technical Architecture

System Architecture

Frontend: Next.js for a modern, fast user interface that works on all devices.

Backend: Node.js to handle server-side logic, user authentication, and item matching.

Database: Cloud Database (PostgreSQL) to store item reports, user data, and matching records

with real-time updates.

Database Schema (Simplified)

Users: id, name, email (MTU unique), role (Student/Admin).

Items: id, reporter_id, type (Lost/Found), category, description_public, description_private, location_tag, status, image_url.

Claims: id, item_id, seeker_id, verification_text, proof_url, status (Pending/Approved).

7. 🎯 Frontend & UX Guidelines

Mobile-First Design: Optimized for students reporting items while moving between classes.

Redaction Logic: Public views must hide "verification-critical" info (e.g., specific serial numbers or unique identifiers).

Accessibility: Simple, clean interface to ensure ease of use during high-stress situations.

8. 🔐 Security, Risks & Constraints

Honesty Constraint: The system only works if people choose to report what they find.

No Delivery: The portal identifies item location, but physical retrieval is required.

Internet Dependency: Users must be online to access the portal or receive match notifications.

Verification Logic: To prevent "Zero Wrongful Claims", owners must provide details not visible in the public post.

9. 📅 Roadmap & Methodology

Phase 1: Research (Step 1)

Study current student reporting habits and identify system requirements.

Phase 2: Build (Step 2)

Develop the web platform, database, and matching features.

Phase 3: Test & Launch (Step 3)

Test with real users, fix bugs, and launch campus-wide